MITTAG-LEFFLER FOR THE COHOMOLOGY OF AN INVERSE SYSTEM, IN THE FORM USING THE IMAGES OF THE BOUNDARY MAPS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for the sharper statement, including the graded compatibility of the boundary maps and the final stable-image computation.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Mittag-Leffler condition and its stable-image form. Same chapter, the Mittag-Leffler condition for an inverse system of modules, in the equivalent stable-image form stated in the footnote of the lemma. Supplies the conclusion of the statement.
- Ascending chain condition on a graded module over the ring of the powers of I. Same chapter, the graded module structure of the direct sum of cohomology groups over the direct sum of the I^n/I^{n+1}, and the ascending chain condition hypothesis. Supplies the finite generation used in the proof.
- Long exact cohomology sequence and the graded compatibility of the boundary maps. Same chapter, the boundary maps attached to the short exact sequences cutting out F_n and the compatibility delta_{n+m}(fs) = f delta_n(s). Supplies the graded submodule whose finite generation drives the argument.

Source reference. Original lemma 0GYQ: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 4731–4731. Navigation pointer invoked at proof line 4814; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:sites-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let I be an ideal of a ring A. Let $\mathcal{C}$ be a site. Let*

$$\ldots \to \mathcal{F}_3 \to \mathcal{F}_2 \to \mathcal{F}_1$$

be an inverse system of A-modules on $\mathcal{C}$ such that $\mathcal{F}_n = \mathcal{F}_{n+1}/I^n\mathcal{F}_{n+1}$. Let $p \geq 0$. Given n define

$$N_n = \bigcap\nolimits_{m \geq n} \mathrm{Im}\left(H^{p+1}(\mathcal{C}, I^n\mathcal{F}_{m+1}) \to H^{p+1}(\mathcal{C}, I^n\mathcal{F}_{n+1})\right)$$

If $\bigoplus N_n$ satisfies the ascending chain condition as a graded $\bigoplus_{n\geq 0} I^n/I^{n+1}$-module, then the inverse system $M_n = H^p(\mathcal{C}, \mathcal{F}_n)$ satisfies the Mittag-Leffler condition[1].

Proof. The proof is exactly the same as the proof of Lemma 0GYQ. In fact, the result will follow from the arguments given there as soon as we show that $\bigoplus N_n$ is a graded $\bigoplus_{n\geq 0} I^n/I^{n+1}$-submodule of $\bigoplus H^{p+1}(\mathcal{C}, I^n\mathcal{F}_{n+1})$ and that the boundary maps $\delta_n : M_n \to H^{p+1}(\mathcal{C}, I^n\mathcal{F}_{n+1})$ have image contained in N_n.

Suppose that $\xi \in N_n$ and $f \in I^k$. Choose $m \gg n+k$. Choose $\xi' \in H^{p+1}(\mathcal{C}, I^n\mathcal{F}_{m+1})$ lifting ξ. We consider the diagram

$$\begin{array}{ccccccccc}
0 & \longrightarrow & I^n\mathcal{F}_{m+1} & \longrightarrow & \mathcal{F}_{m+1} & \longrightarrow & \mathcal{F}_n & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f} & & \\
0 & \longrightarrow & I^{n+k}\mathcal{F}_{m+1} & \longrightarrow & \mathcal{F}_{m+1} & \longrightarrow & \mathcal{F}_{n+k} & \longrightarrow & 0
\end{array}$$

constructed as in the proof of Lemma 0GYQ. We get an induced map on cohomology and we see that $f\xi' \in H^{p+1}(\mathcal{C}, I^{n+k}\mathcal{F}_{m+1})$ maps to $f\xi$. Since this is true for all $m \gg n+k$ we see that $f\xi$ is in N_{n+k} as desired.

To see the boundary maps δ_n have image contained in N_n we consider the diagrams

$$\begin{array}{ccccccccc}
0 & \longrightarrow & I^n\mathcal{F}_{m+1} & \longrightarrow & \mathcal{F}_{m+1} & \longrightarrow & \mathcal{F}_n & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & I^n\mathcal{F}_{n+1} & \longrightarrow & \mathcal{F}_{n+1} & \longrightarrow & \mathcal{F}_n & \longrightarrow & 0
\end{array}$$

for $m \geq n$. Looking at the induced maps on cohomology we conclude. □

[1] In fact, there exists a $c \geq 0$ such that $\mathrm{Im}(M_n \to M_{n-c})$ is the stable image for all $n \geq c$.